A Design Application Handbook of **Indirect Biophilic Attributes in Hawaii Hotel Interior Design**

——Comparative Study on Waikiki 4-star hotels

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EDRA55 Portland Individual Presentation

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Direct Experience of Nature	Indirect Experience of Nature	Experience of Space and Place
• Light • Air • Water • Plants • Animals • Weather • Natural landscapes and ecosystems • Fire	• Images of nature • Natural materials • Natural colors • Simulating natural light and air • Naturalistic shapes and forms • Evoking nature • Information richness • Age, change, and the patina of time • Natural geometries • Biomimicry	• Prospect and refuge • Organized complexity • Integration of parts to wholes • Transitional spaces • Mobility and wayfinding • Cultural and ecological attachment to place

https://www.biophilic-design.com/

Kellert Biophilic design 24 attributes

A Design Application Handbook of

Indirect Biophilic Attributes in Hawaii

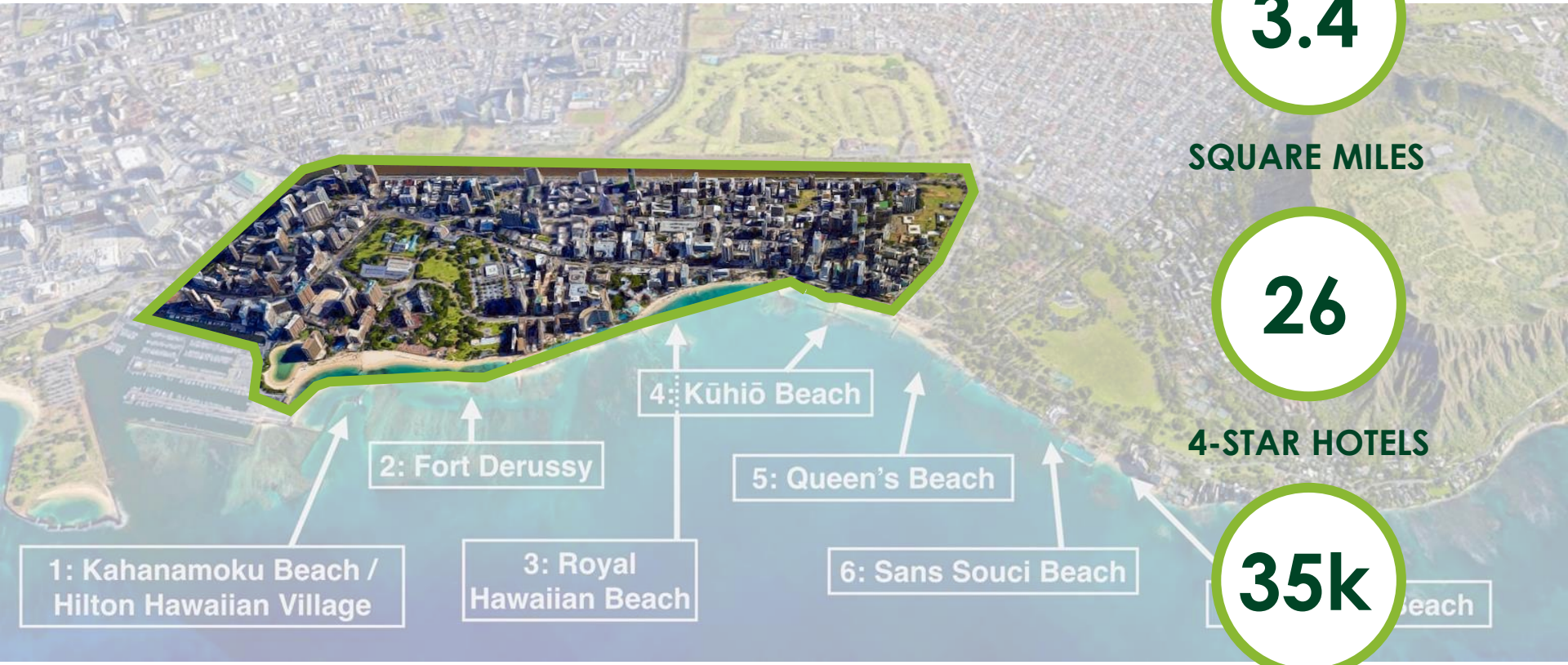
Hotel Interior Design

——Comparative Study on Waikiki 4-star hotels

Accessible Post-occupancy Feedbacks
Highly unified space
Online Photos
Similar decision-making process as customers
……

A Design Application Handbook of **Indirect Biophilic Attributes in Hawaii Hotel Interior Design**

——Comparative Study on Waikiki 4-star hotels



3.4

SQUARE MILES

26

4-STAR HOTELS

35k

ONLINE COMMENTS

A Design Application Handbook of

Indirect Biophilic Attributes in Hawaii Hotel Interior Design

——Comparative Study on Waikiki 4-star hotels

How can we use those “Legos” to improve our design?

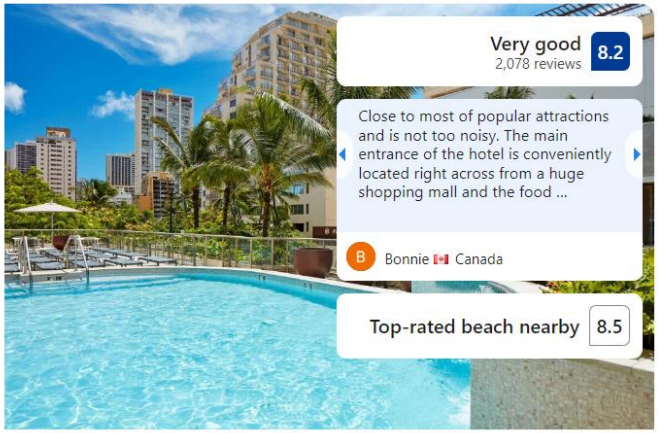
Hilton Garden Inn Waikiki Beach

We Price Match

2330 Kuhio Avenue, Waikiki, Honolulu, HI 96815, United States – **Excellent location - show map**







Very good 8.2
2,078 reviews

Close to most of popular attractions and is not too noisy. The main entrance of the hotel is conveniently located right across from a huge shopping mall and the food ...

B Bonnie Canada

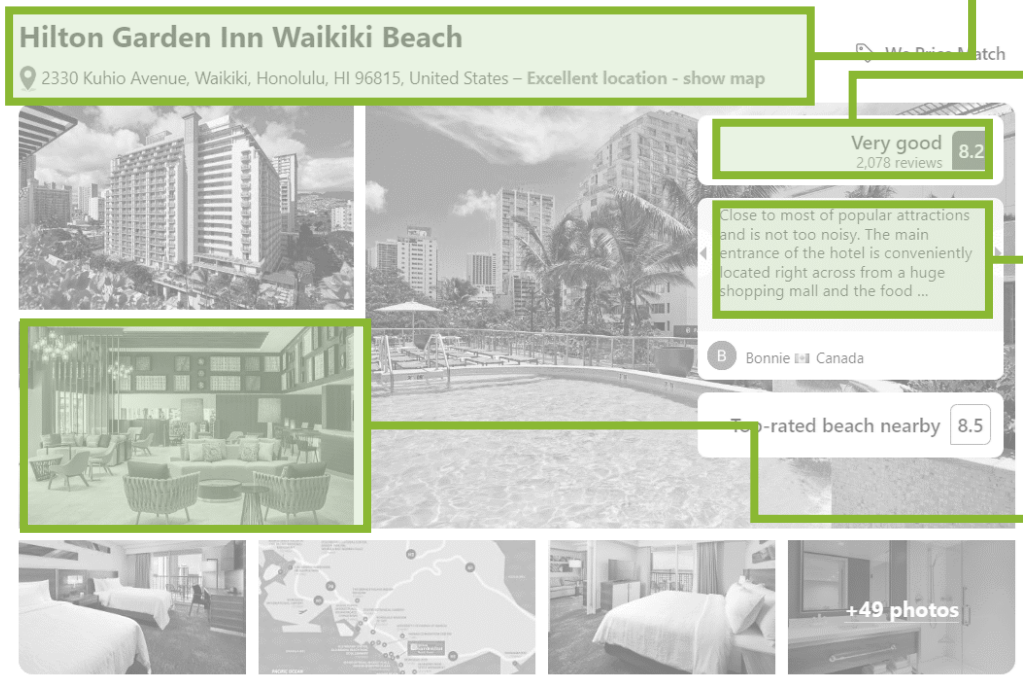
Top-rated beach nearby 8.5











Basic information

- Location
- Name
- Rating
- Star/price positioning
- Average price
- Click time

Rating

- Overall
- Comfort
- Facilities
- Value for money

Comments

- Number
- Details / key words

Photos

- Lobby
- Main entrance
- Outdoor recreation area
- Typical guest room
- Indoor public activity area
- Surroundings

Basic Info

Ratings

Photos

Comments

Basic Info

Ratings

Consumption willingness
Comfort evaluation
Quality Assessment

Photos

Indirect
Biophilic
Attributes

Comments

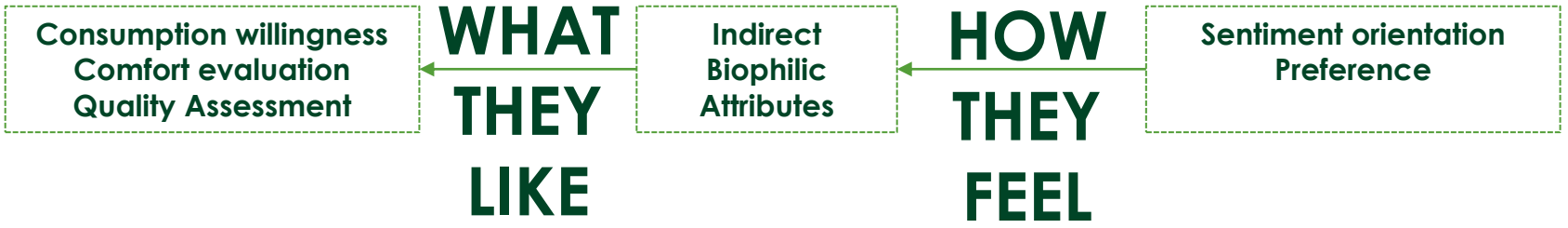
Sentiment orientation
Preference

Basic Info

Ratings

Photos

Comments

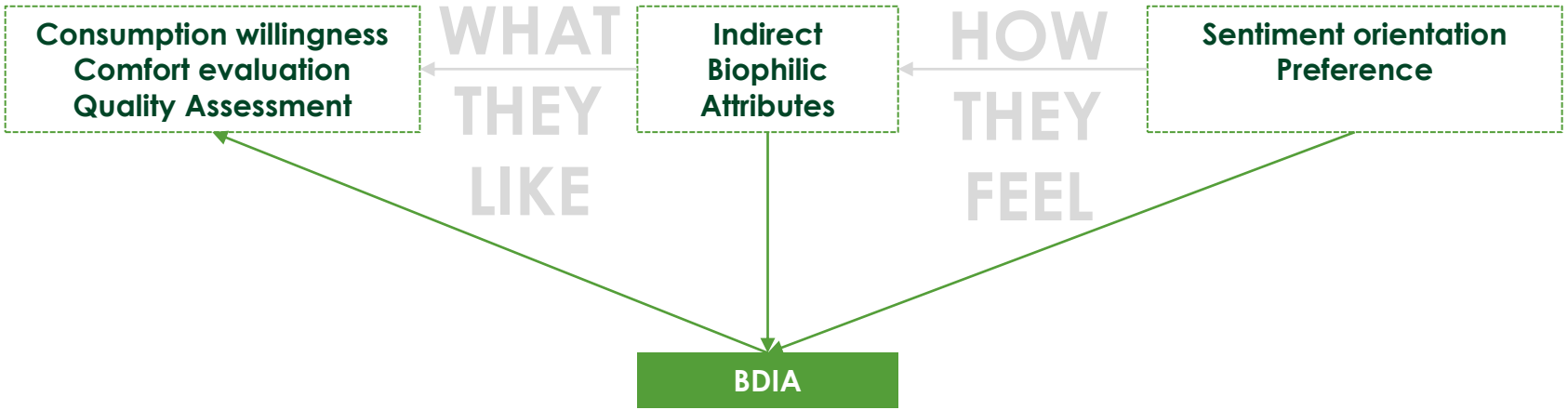


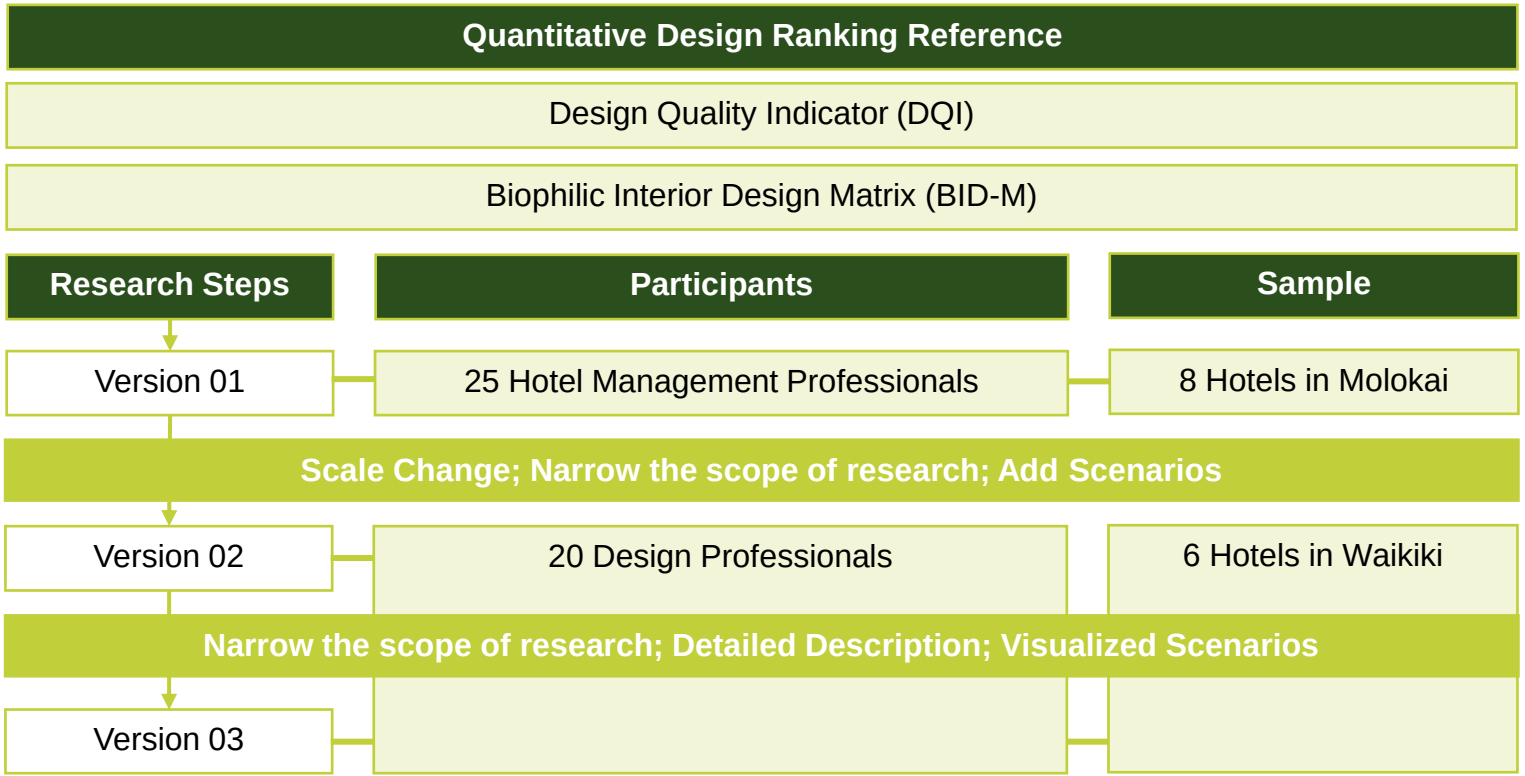
Basic Info

Ratings

Photos

Comments




Quantitative Design Ranking Reference
Design Quality Indicator (DQI)
Biophilic Interior Design Matrix (BID-M)
Research Steps
Participants
Sample
Version 01
25 Hotel Management Professionals
8 Hotels in Molokai
Scale Change; Narrow the scope of research; Add Scenarios
Version 02
20 Design Professionals
6 Hotels in Waikiki
Narrow the scope of research; Detailed Description; Visualized Scenarios
Version 03

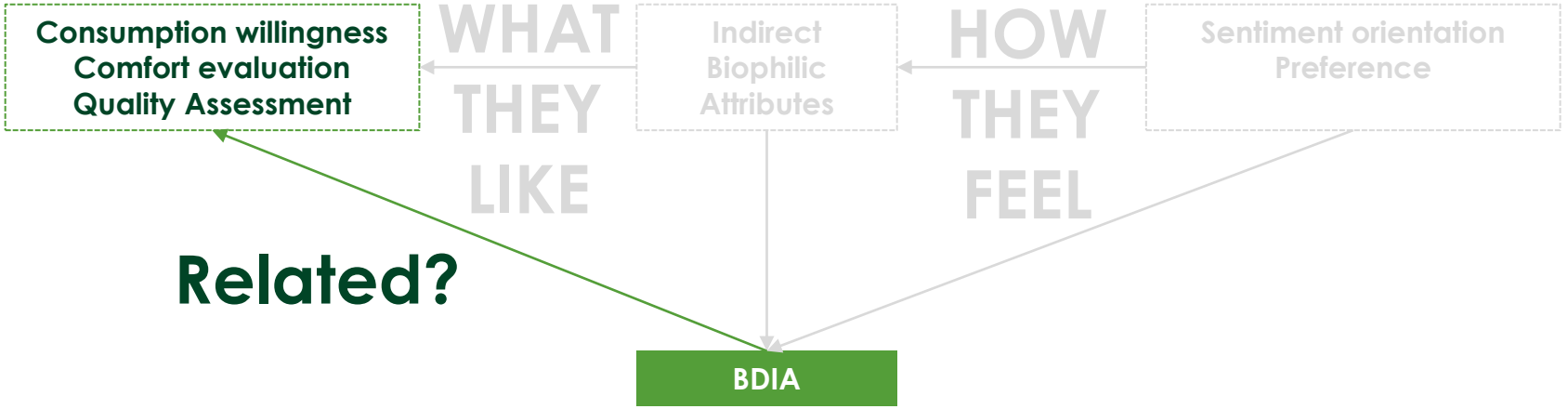
Num.	BDIA	Guestroom	Lobby	Restaurant
A01	Wall paintings or decorative paintings contain direct natural scenes (such as landscape photos).	1	1	
A02	Carpets, pillows, curtains and other fabrics contain direct natural scenes (such as cartoon cat pillows).	1	1	
A03	Use decorations with natural forms (such as ceramic rockeries).	1		
B01	Background walls, columns and other hard wall decorations are made of natural materials.		1	1
B02	Window frames, sun visors and balcony railings are made of natural materials.			1
B03	The ceiling is made of natural materials.		1	1
B04	The floors are made of natural materials.	1		
B05	The floors are made of natural materials.	1	1	1
B06	Decorations are made of natural materials.	1	1	1
C01	The hard decoration of the background wall, pillars and other walls are in natural colors.	1	1	1
C02	Window frames, sun visors and balcony railings are in natural colours.			1
C03	The ceilings are in natural colours.		1	1
C04	The floors are in natural colours.	1	1	1
C05	Furniture is in natural colours.	1	1	1
C06	Decorations are in natural colors.	1	1	1
D01	Adjustable sash windows allow for natural ventilation.	1	1	1
D02	Adjustable sun visors or blinds.		1	1
D03	Use indirect ambient light.			
D04	Lamp with soft lighting effect.	1		1
E01	The furniture adopts irregular natural forms.	1	1	
E02	Hard decoration (walls, ceilings, etc.) adopts irregular natural forms.			
E03	The decorations adopt irregular natural forms.	1	1	
F01	There are nostalgic elements in the decoration style (such as 90s posters).	1	1	1
F02	The decorative style includes nostalgic craftsmanship (such as bamboo weaving and hand binding).	1	1	1
F03	Use recycled materials, furniture or decor.	1	1	
F04	Use materials that will show natural weathering or aging (such as copper).	1	1	1
G01	The furniture adopts abstract and geometric patterns that imitate nature.	1	1	1

Basic Info

Ratings

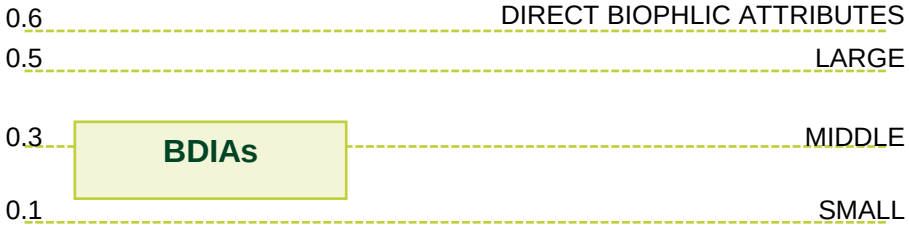
Photos

Comments



Pearson Correlation					
	Overall Score	Facilities	Comfort	Value of Money	Average Price
A01	**0.317****	**0.284****	**0.314****	0.107	**0.202***
A02	0.153	**0.158***	0.092	-0.034	0.139
A03	**0.192***	0.096	**0.205***	0.082	0.008
B01	**0.323****	**0.304****	**0.337****	**0.201***	-0.076
B02	0.022	0.006	0.033	-0.038	0.06
B03	**0.195***	**0.227****	**0.216****	**0.313****	**-0.205***
B04	0.044	0.143	0.106	-0.133	**0.235****
B05	**0.201***	**0.240****	**0.243****	0.071	0.105
B06	**0.239****	**0.275****	**0.307****	**0.161***	**0.169***
C01	**0.217****	**0.240****	**0.256****	0.117	0.036
C02	0.131	**0.162***	0.156	0.103	0.125
C03	0.147	**0.221****	**0.176***	0.143	-0.001
C04	0.138	**0.202***	**0.169***	-0.061	**0.254****
C05	**0.227****	**0.202***	**0.243****	0.157	-0.054
C06	**0.309****	**0.292****	**0.342****	**0.236****	-0.05
D01	**0.161***	0.11	0.142	0.09	0.069
D02	-0.018	-0.053	0.014	-0.054	0.045
D03	**0.245****	**0.304****	**0.281****	0.137	**0.264****
D04	**0.204***	**0.213****	**0.224****	0	**0.210****
E01	0.111	0.180*	0.156	0.230**	-0.131
E02	0.022	0.016	0.045	-0.131	0.022
E03	**0.172***	**0.173***	**0.192***	**0.237****	-0.041
F01	0.131	0.101	0.079	0.051	0.071
F02	**0.348****	**0.296****	**0.369****	**0.388****	-0.064
F03	0.095	0	0.012	**0.211****	-0.023
F04	0.08	0.064	0.029	**0.198***	-0.109
G01	**0.441****	**0.414****	**0.435****	**0.253****	**0.182***
G02	**0.259****	**0.281****	**0.310****	0.087	0.076
G03	**0.346****	**0.389****	**0.358****	0.135	**0.248****

* p<0.05 ** p<0.01



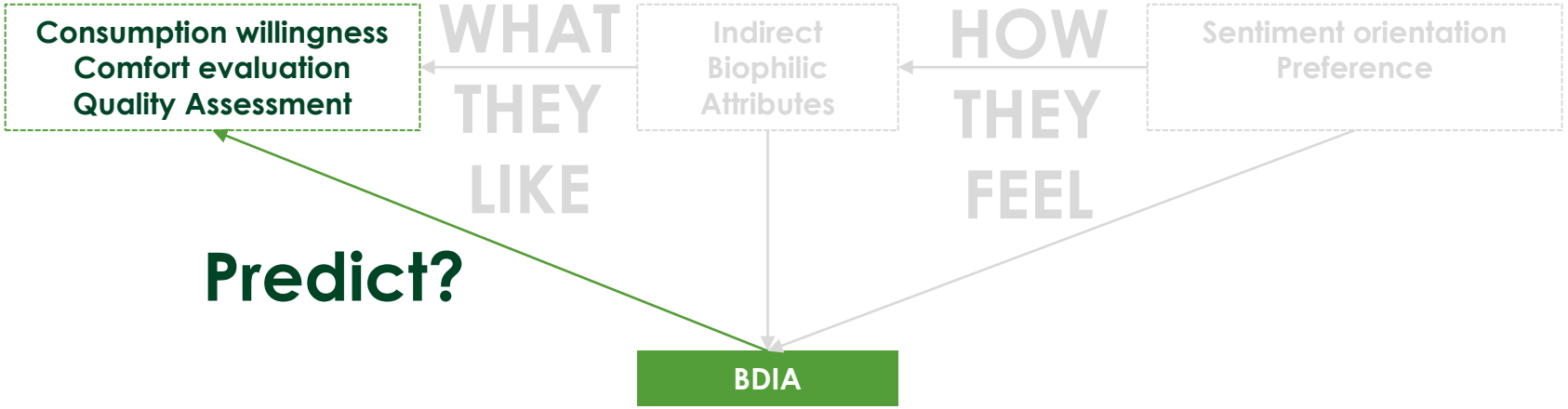
The results of the correlation analysis showed that **BDIA is significantly related to rankings in booking.com.**

Basic Info

Ratings

Photos

Comments

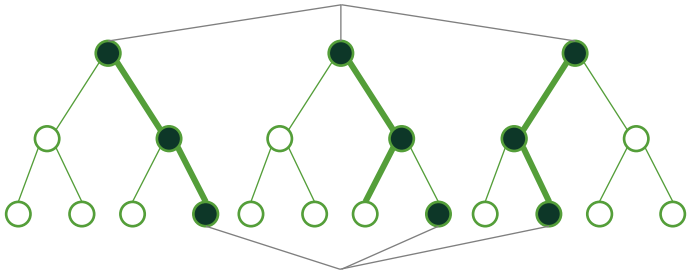


Random Forest Algorithm

The Random Forest algorithm is a sophisticated machine learning technique that operates on the principle of ensemble learning.

.

- Utilize Random Forest algorithm for accurate and robust decision-making in complex data scenarios.



FINAL RESULT

Model summary table		
Name	parameter names	parameter settings
Model parameter settings	Data preprocessing	None
	Training set proportion	0.9
	Number of decision trees	100
	Node Split Criteria	squared_error
	Minimum number of samples for node splitting	2
	Minimum number of samples for leaf nodes	1
	tree maximum depth	None
	Maximum number of features	auto
	Whether there is sampling with replacement	Yes
	Whether to perform out-of-bag data testing	Yes

Model evaluation result		
Index	Training	Testing
R squared value	0.961	0.732
Mean absolute error MAE	0.057	0.218
Mean square error MSE	0.007	0.066
root mean square error RMSE	0.085	0.257
Median absolute error MAD	0.033	0.247
Mean Absolute Percent Error MAPE	0.01	0.004
Explained variance score EVS	0.961	0.733
Root mean square log error MSLE	0	0.001

Attributes		
A01	Nature Image Finishing	0.056
A03	Nature Image Decorations	0.03
B01	Natural materials Walls	0.05
B03	Natural materials Ceiling	0.06
B04	Natural materials Floor	0.054
B06	Natural materials Decorations	0.103
C01	Natural colors Walls	0.045
C03	Natural colors Ceiling	0.052
D01	Adjustable Windows	0.05
D03	Indirect Lighting	0.061
E03	Naturalistic shapes and forms Decorations	0.07
F01	Nostalgic Decorations	0.03
F02	Nostalgic Craftsmanship	0.048
G01	Natural geometries Furnitures	0.18
G02	Natural geometries Finishing	0.068
G03	Natural geometries Furnishing	0.042

The proportion of the training set is adjusted to 0.9, and the calculation is carried out when the data is normalized. The specific value is set as follows. The model training and testing results are matched, and the interpretation rate is high.

BDIA-Index = SUM(BDIA*Corresponding weight)

- The BDIA-Index assigned according to the weight of Overall Score has strong correlation with all kinds of scores.

Booking Overall Score=7.304 + 1.911*BDIA-Index

- The R-square value of the model is 0.504, which means that the BDIA-Index can explain **50.4%** of the change in the Overall Score.
- It was found that the model passed the F test (F=156.505, p=0.000< 0.05), that is, BDIA-Index must have an impact on the Overall Score.
- The regression coefficient of BDIA-Index was 1.911(t=12.510, p=0.000< 0.01), which means that BDIA-Index will have a significant positive impact on the Overall Score.

Pearson Correlation								
	Booking Overall Score	Booking Facilities Score	Booking Comfort	Booking Value for money	Google Score	Expedia Amenities Score	Expedia Property conditions & facilities Score	Expedia Overall Score
Booking Score Prediction	**0.883***	**0.859***	**0.840***	**0.649**	**0.614**	**0.743**	**0.796**	**0.823***
* p<0.05 ** p<0.01								

Parameter Estimates (n=156)							
	Unstandardized Coefficients		Standardized Coefficients	t	p	Collinearity diagnosis	
	B	Std. Error	Beta			VIF	Tolerance
Constant	7.292	0.136	-	53.757	0.000**	-	-
BDIA-Index	1.854	0.218	0.698	8.503	0.000**	1	1
R 2	0.488						
Adj R 2	0.481						
F	F (1,76)=72.309,p=0.000						
D-W Value	0.623						
Dependent Variable: Overall Score							
* p<0.05 ** p<0.01							

BDIA-Index = SUM(BDIA*Corresponding weight)

Booking Overall Score=7.304 + 1.911*BDIA-Index

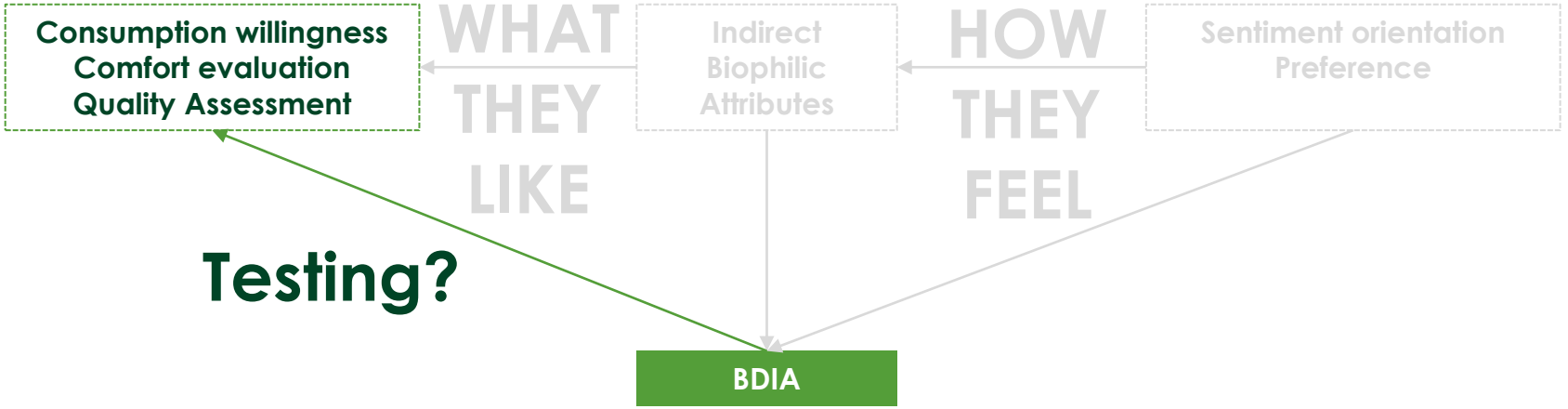
- The positive correlation between BDIA scores and higher ratings on these platforms not only indicates that a greater use of BDIA elements can significantly improve online scores and customer reviews but also suggests an increase in perceived value, comfort, and facility quality.
- The perception of value for money did not show a significant correlation with BDIA-Index.
- Hotels with higher BDIA-Index tend to rank higher in listings, which further validates the index as a useful tool for enhancing hotel competitiveness.

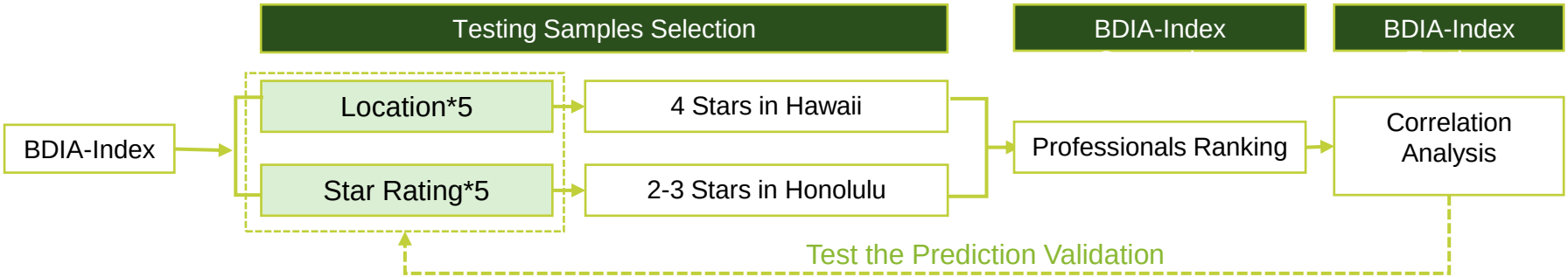
Basic Info

Ratings

Photos

Comments

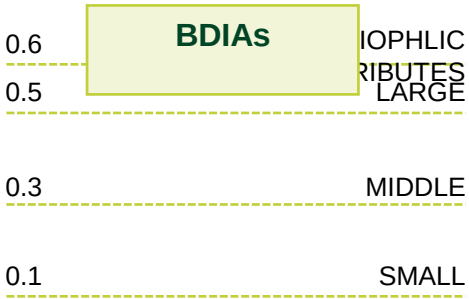



Testing Samples Selection
BDIA-Index
BDIA-Index
BDIA-Index
Location*5
Star Rating*5
4 Stars in Hawaii
2-3 Stars in Honolulu
Professionals Ranking
Correlation Analysis
Test the Prediction Validation

Correlation Analysis Results

	BOOKING RANKING PREDICTION	BOOKING SCORE PREDICTION	BDIA-INDEX
GOOGLE SCORE	-0.566	0.416	0.416
EXPEDIA SCORE	-0.583	0.659	0.659
OVERALL SCORE RANKING	**0.685***	**-0.679***	**-0.679***
OVERALL SCORE	-0.5	0.582	0.582
FACILITIES	-0.572	**0.706***	**0.706***
COMFORT	**-0.659***	**0.687***	**0.687***
VALUE OF MONEY	0.153	0.059	0.059
AVERAGE PRICE	**-0.689***	**0.673***	**0.673***

BDIA-index is an efficient and significant predictor of perceived quality for hotels of different levels and geographical locations in Hawaii.

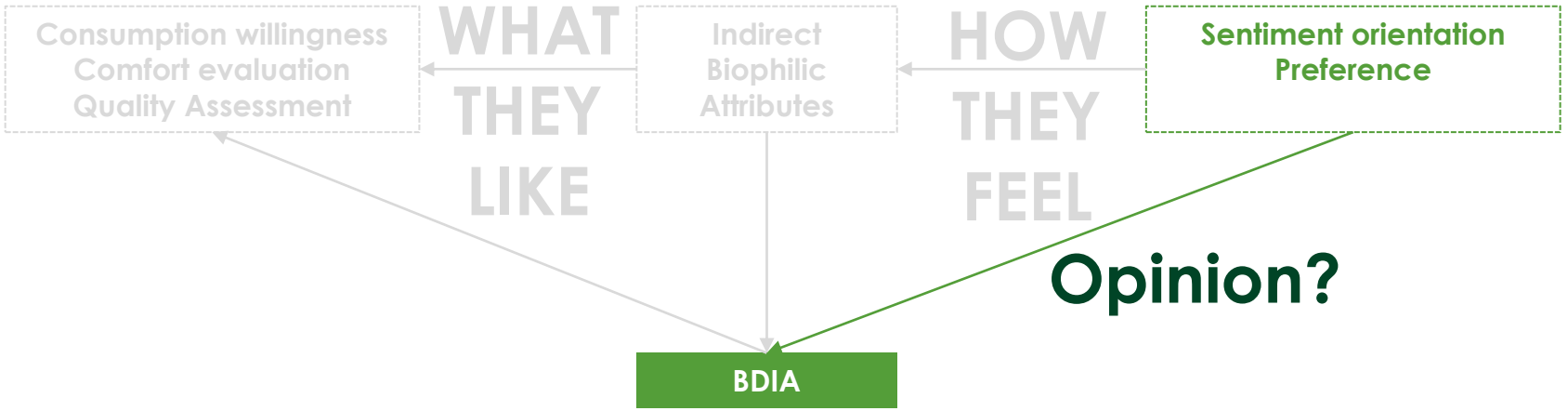


Basic Info

Ratings

Photos

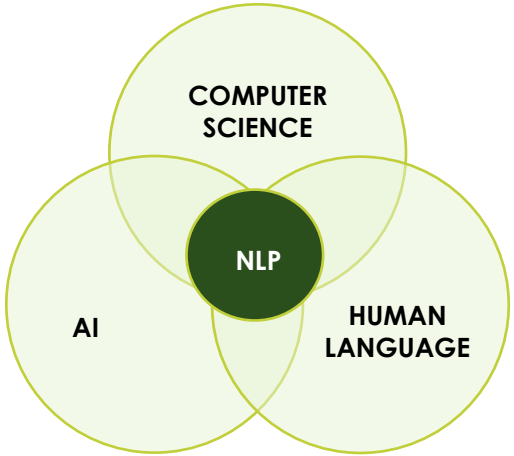
Comments

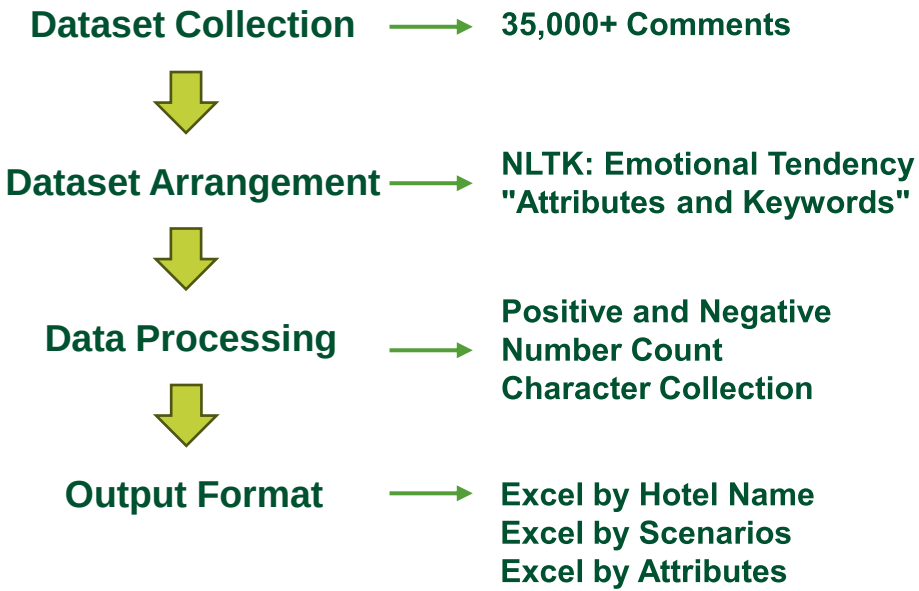


Natural Language Processing

Natural Language Processing (NLP) is a pivotal domain within artificial intelligence that focuses on the interaction between computers and human (natural) languages.

- Analyze textual feedback to understand customer sentiments and preferences.




Dataset Collection
35,000+ Comments
Dataset Arrangement
NLTK: Emotional Tendency
"Attributes and Keywords"
Data Processing
Positive and Negative
Number Count
Character Collection
Output Format
Excel by Hotel Name
Excel by Scenarios
Excel by Attributes



Vintage Style

- significantly correlation with positive rankings
- fewer occurrences (Mean=0.34

a new design potential opportunity and can be more actively adopted in future designs.



Local Identity

Local Identity is related to higher Booking scores, higher Quantitative Feeling And Evaluation, and more positive consumption decisions.



Nature Image / Geometry

- strong positive predictors
- a potential design standard for local high-end hotels

can continue to be used in future designs and can refer to existing design plans



Natural colors and materials

- negative correlation
- associate with a lower grade or outdated

Naturalistic shapes and forms Finishing is either associated with negative outcome scores or not associated with results in all analyses, and should be avoided in design as much as possible.



Hawaii / Vintage Style

- Local, retro, or nostalgic style, was related to more positive sentiment.
- Fewer occurrences (Mean=0.34), which is not a mainstream design style.



Balcony

- More positive (403) than negative (122)
- Noise problems (88)
- May be related to thermal discomfort
- Allowing breezes and fresh air
- Direct sunlight can cause overheating



Music / Smell

- Playing music, enhancing atmosphere, fragrance and helping to increase positive evaluation
- increasing good tactile experience has a slight positive effect.
- White noise and bad odors can have a negative effect.

Negative



Natural Material
Flooring



Natural material
Ceiling



Natural Color
Flooring

Positive



Recycled Furniture &
decorations



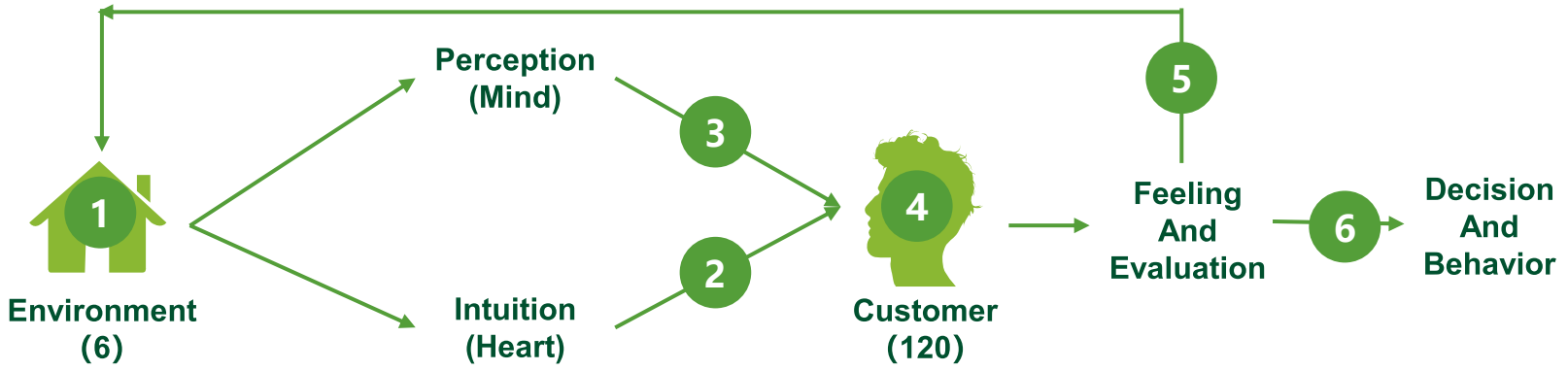
Craftsmanship



Decorations
intimitate nature

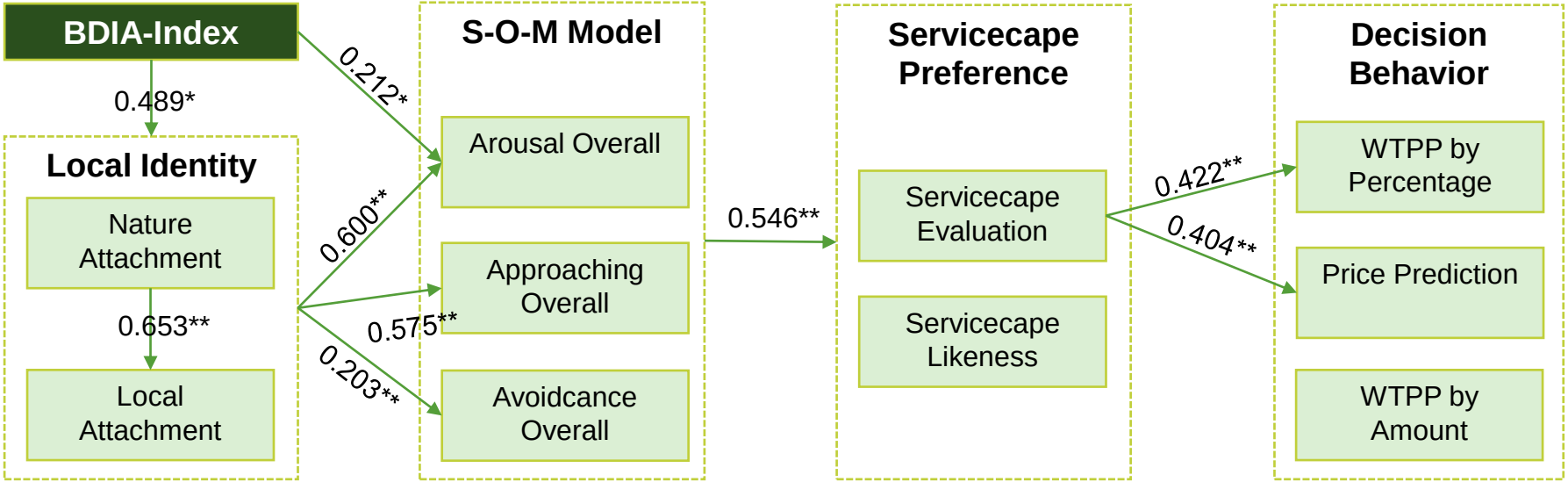
The two–way path does exist.

But how and why?



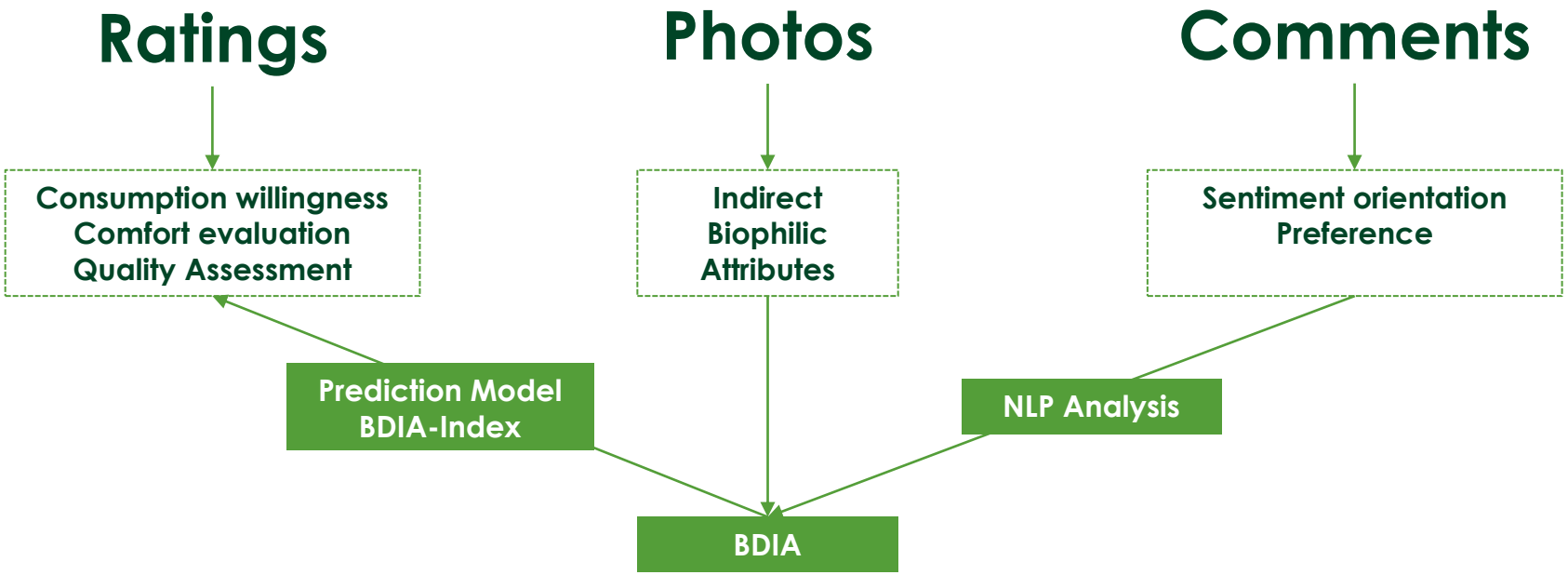
Measurements	Theory	No	Aim
Customer	**Prequalification**	5	to ensure the customers are familiar with hotel staying and online booking
	Demographic Information	/	Eliminate between-group difference
Intuition	**S-O-R Model**	10	to ensure the customers are familiar with hotel staying and online booking
Perception	**Local Identity**	10	to measure the degree of “Nature” and “Hawaii” in the interior design
Feeling And Evaluation	**Servicescape Evaluations**	9	to measure customer preference, quality perception, and satisfaction with hotel interior design
Decision And Behavior	**Feeling And Evaluation**	8	measure people’s Predetermined intention and WTPP/willingness to pay a premium

Path Analysis

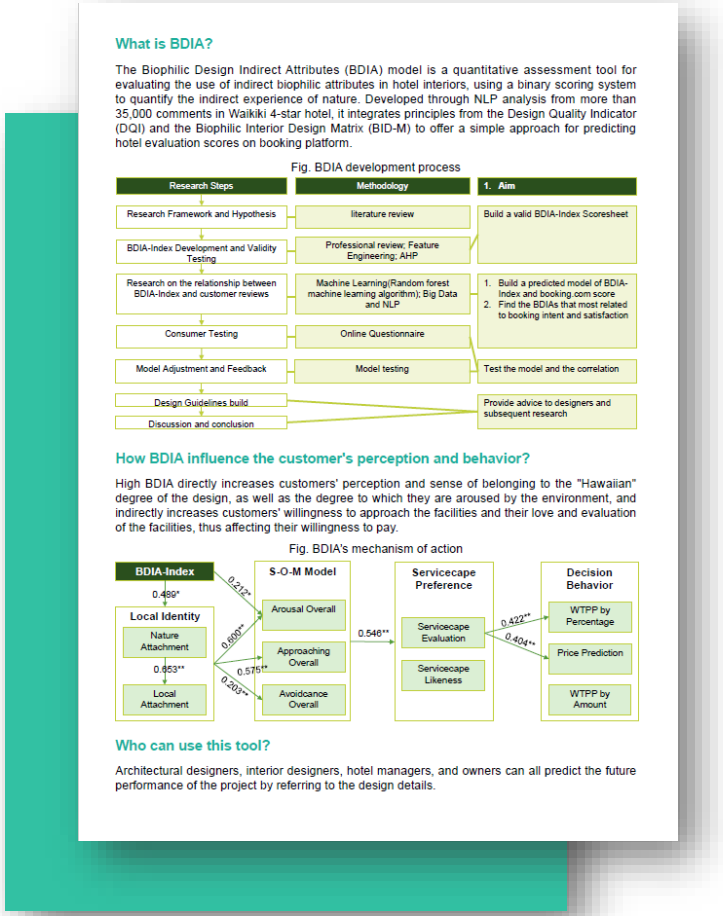


Through horizontal comparison of the Standardized Estimate, t-Statistic and p-Value results of other research path analysis, it was found that the effect size of BDIA-Index on improving Local Identity and other design perception factors was between medium and large. Combined with the goodness-of-fit statistics of the model showed that the structural model fit the data very well, it can be considered that this model better reflects the impact mechanism of BDIA on real customer decision-making and Impact size.

Basic Info


Ratings
Photos
Comments
Consumption willingness
Comfort evaluation
Quality Assessment
Indirect
Biophilic
Attributes
Sentiment orientation
Preference
Prediction Model
BDIA-Index
NLP Analysis
BDIA

A Design Application Handbook



What is BDIA?

The Biophilic Design Indirect Attributes (BDIA) model is a quantitative assessment tool for evaluating the use of indirect biophilic attributes in hotel interiors, using a binary scoring system to quantify the indirect experience of nature. Developed through NLP analysis from more than 35,000 comments in Waikiki 4-star hotel, it integrates principles from the Design Quality Indicator (DQI) and the Biophilic Interior Design Matrix (BID-M) to offer a simple approach for predicting hotel evaluation scores on booking platform.

Fig. BDIA development process

How BDIA influence the customer's perception and behavior?

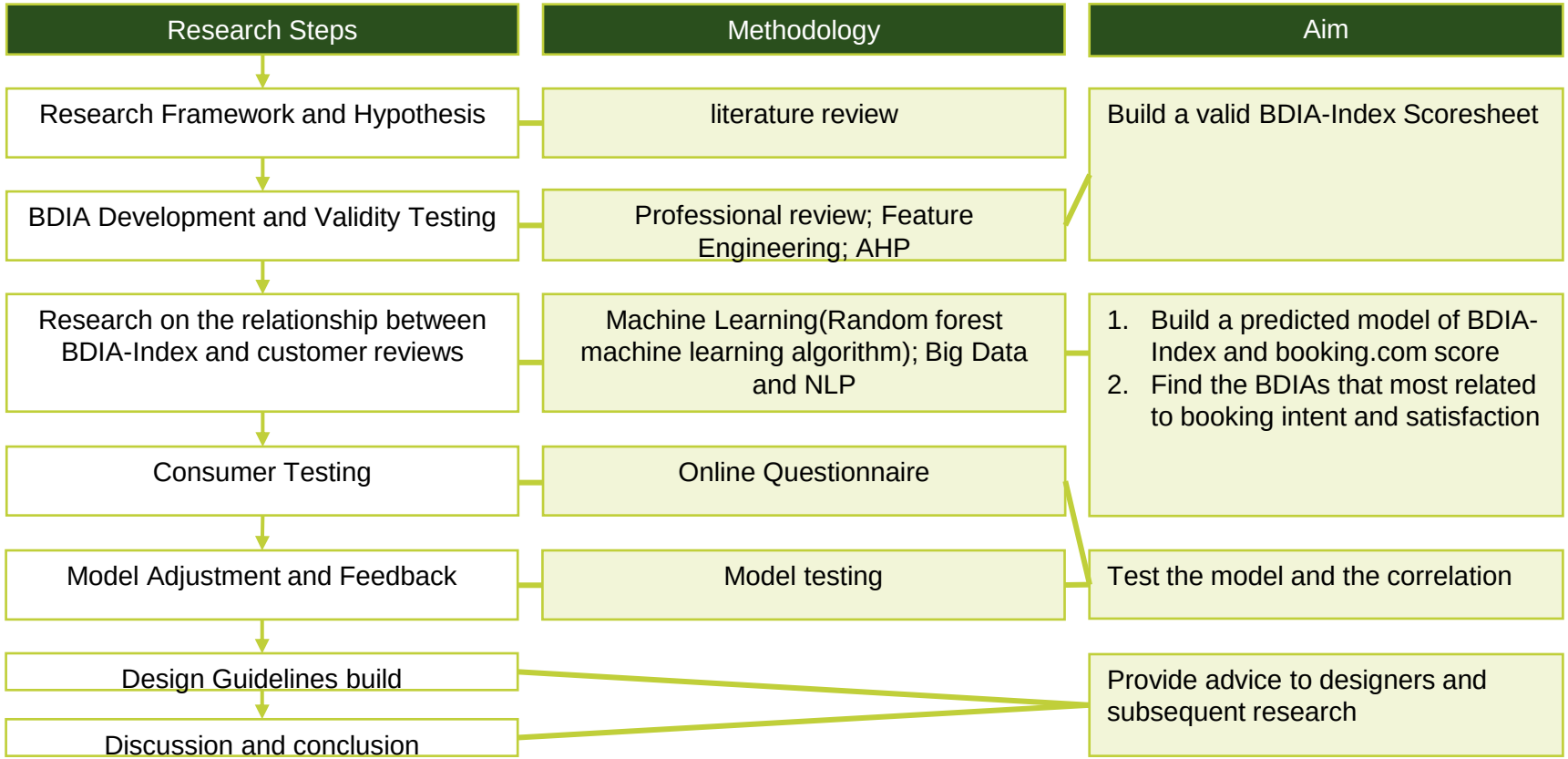
High BDIA directly increases customers' perception and sense of belonging to the "Hawaiian" degree of the design, as well as the degree to which they are aroused by the environment, and indirectly increases customers' willingness to approach the facilities and their love and evaluation of the facilities, thus affecting their willingness to pay.

Fig. BDIA's mechanism of action

Who can use this tool?

Architectural designers, interior designers, hotel managers, and owners can all predict the future performance of the project by referring to the design details.

SUMMARY


Research Steps
Methodology
Aim
Research Framework and Hypothesis
literature review
Build a valid BDIA-Index Scoresheet
BDIA Development and Validity Testing
Professional review; Feature Engineering; AHP
Research on the relationship between BDIA-Index and customer reviews
Machine Learning(Random forest machine learning algorithm); Big Data and NLP
1. Build a predicted model of BDIA-Index and booking.com score
2. Find the BDIAs that most related to booking intent and satisfaction
Consumer Testing
Online Questionnaire
Model Adjustment and Feedback
Model testing
Test the model and the correlation
Design Guidelines build
Discussion and conclusion
Provide advice to designers and subsequent research

Thank you!

Chunya Wu

University of Hawaii at Manoa; ZGF Architects

EDRA55 Portland Individual Presentation

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